

Performance & Unit Testing

Testing and Verification Report - Rising Waters

1. Automated Integration Testing

We configured a python test script ``test_vercel_app.py`` utilizing the Flask test client to verify routes, parameters, and redirects without spinning up a live server.

Test Matrix Summary

All 9 test scenarios passed successfully in 0.057 seconds:

Test ID	Function tested	Input Parameters	Expected Outcome	Status
01	GET <code>`/`</code> (Home)	None	Status 200, contains "Rising Waters"	PASSED
02	GET <code>`/Predict`</code> (Form)	None	Status 200, contains "Cloud Cover"	PASSED
03	POST <code>`/predict`</code> (Low-risk)	Cloud: 5%, Annual: 1200mm	Redirects to <code>`/no_chance`</code> with prob: 99.9%	PASSED
04	POST <code>`/predict`</code> (High-risk)	Cloud: 90%, Annual: 4500mm	Redirects to <code>`/chance`</code> with prob: 99.9%	PASSED
05	POST <code>`/predict`</code> (Negative Value)	Cloud: 50%, Annual: -1000mm	Redirects to <code>`/index`</code> , flashes message	PASSED
06	POST <code>`/predict`</code> (Cloud Range)	Cloud: 105%, Annual: 2000mm	Redirects to <code>`/index`</code> , flashes message	PASSED
07	POST <code>`/predict`</code> (Logical Sum)	Annual: 1000mm, Seasonal: 1200mm	Redirects to <code>`/index`</code> , flashes message	PASSED
08	GET <code>`/chance`</code> (Warning)	prob: 98.55%, inputs: high-risk	Status 200, displays warning	PASSED
09	GET <code>`/no_chance`</code> (Safe)	prob: 92.10%, inputs: low-risk	Status 200, displays safe and sound	PASSED

2. Model Performance Assessment

Evaluated on the 25% test split (random state 10):

- Accuracy Score: 96.55% (28 correct classifications out of 29 samples).
- F1-Score: 96.55%
- Precision / Recall: 96.55%

Confusion Matrix Outputs

- True Negatives (No Flood): 12
- False Negatives: 1
- True Positives (Flood Risk): 16
- False Positives: 0
- *Generalization*: The model displays 0% false positives, ensuring warning alerts are only triggered under genuine hazard parameters.